MAKING A LOCALLY CONSTANT F-L SHEAF CONSTANT AFTER A FINITE ETALE COVER OF DEGREE PRIME TO L

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; the covering, its degree and the filtration are all produced inside the counted span.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Finite type locally constant sheaves of F_l-vector spaces and the associated finite representations of the fundamental group. Same chapter, the classification of finite type locally constant sheaves of F_l-vector spaces by continuous representations of the etale fundamental group on finite dimensional F_l-vector spaces. Supplies the translation of the problem into a statement about a finite group representation.
- Existence of finite etale covers trivialising a finite locally constant sheaf. Same chapter, the description of a locally constant sheaf as constant over the members of a finite etale covering. Supplies the covering over which the sheaf becomes constant.
- Representations of a finite group of order prime to l and their lifts. Same chapter together with the group-theoretic input used by the author: over a finite group of order prime to l a finite dimensional representation in characteristic l is entirely decomposable into characters, so after a suitable cover the filtration by trivial quotients becomes visible. Supplies the argument producing the filtration with quotients Z/l.

Source reference. Original lemma 0DV5: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 10118–10118. Navigation pointer invoked at proof

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

line 10426; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03SF: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 10437–10437. Navigation pointer invoked at proof line 10437; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original remark 0DV6: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 10294–10294. Navigation pointer invoked at proof line 10446; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let S be a connected scheme. Let ℓ be a prime number. Let $\mathcal{F}$ be a finite type, locally constant sheaf of $\mathbf{F}_\ell$-vector spaces on $S_{étale}$. Then there exists a finite étale morphism $f : T \to S$ of degree prime to ℓ such that $f^{-1}\mathcal{F}$ has a finite filtration whose successive quotients are $\underline{\mathbf{Z}/\ell\mathbf{Z}}_T$.*

Proof. Choose a geometric point $\overline{s}$ of S. Via the equivalence of Lemma 0DV5 the sheaf $\mathcal{F}$ corresponds to a finite dimensional $\mathbf{F}_\ell$-vector space V with a continuous $\pi_1(S, \overline{s})$-action. Let $G \subset \mathrm{Aut}(V)$ be the image of the homomorphism $\rho : \pi_1(S, \overline{s}) \to \mathrm{Aut}(V)$ giving the action. Observe that G is finite. The surjective continuous homomorphism $\overline{\rho} : \pi_1(S, \overline{s}) \to G$ corresponds to a Galois object $Y \to S$ of $F\acute{E}t_S$ with automorphism group $G = \mathrm{Aut}(Y/S)$, see Fundamental Groups, Section 03SF. Let $H \subset G$ be an ℓ-Sylow subgroup. We claim that $T = Y/H \to S$ works. Namely, let $\overline{t} \in T$ be a geometric point over $\overline{s}$. The image of $\pi_1(T, \overline{t}) \to \pi_1(S, \overline{s})$ is $(\overline{\rho})^{-1}(H)$ as follows from the functorial nature of fundamental groups. Hence the action of $\pi_1(T, \overline{t})$ on V corresponding to $f^{-1}\mathcal{F}$ is through the map $\pi_1(T, \overline{t}) \to H$, see Remark 0DV6. As H is a finite ℓ-group, the irreducible constituents of the representation $\rho|_{\pi_1(T,\overline{t})}$ are each trivial of rank 1 (this is a simple lemma on representation theory of finite groups; insert future reference here). Via the equivalence of Lemma 0DV5 this means $f^{-1}\mathcal{F}$ is a successive extension of constant sheaves with value $\underline{\mathbf{Z}/\ell\mathbf{Z}}_T$. Moreover the degree of $T = Y/H \to S$ is prime to ℓ as it is equal to the index of H in G. □